Yohan Choi

Hardware Engineer | Seoul, South Korea | johannchoi2002@gmail.com

Experience

EMTOMEGA May 2022 – Present

Hardware Engineer (Assistant Manager), Device Development Team *Seoul, South Korea*

- **TRINITY** (in-house 3-channel QHD+FHD+FHD dashcam): Designed new power and interface circuits for the company's first i.Catch CR5 SoC, supported SW bring-up, and carried the product through reliability testing and KC certification into mass production (15K+ units)
- Resolved an unexpected TRINITY thermal issue by calculating and measuring DC-DC and LDO losses, replacing high-loss LDOs with buck converters, and introducing graphite sheets (new to the company), thermal pads, and heatsinks with the mechanical team
- **OEM Maker-Option Dashcam** (Japanese automaker): Designed new circuits for a wireless front/rear system with time sync and a standalone BLE module that keeps the units talking in sleep mode; verified the dual-MCU-per-unit architecture
- Built a dedicated harness between the main unit and camera module to pass EMC, and ran automaker-standard reliability and EMC testing; now in mass production with a 10K units/year target
- **2-Channel Dashcam Series** (Japanese aftermarket brand, 1M+ units sold including private-brand models): Redesigned peripheral circuits and PCB across derivative models for GPS, NAND flash, backlight driver, MCU, audio codec, and image sensor changes plus further miniaturization; now handling an image sensor change driven by a Sony supply shortage (SOP Jan. 2027)
- Ran CISPR 25 Class 5 radiated and conducted emission tests at prototype stage; traced a failure of the customer's mandatory Class 5 limit to specific parts and layout, then resolved it with shield cans and added filtering
- Validated hardware through ESD and JASO-based environmental reliability testing; fed EMS defect data and DFM requests back into the design with the production team, and handled field issues after launch
- Built and deployed in-house tools with Claude Code to automate repetitive hardware tasks, and applied AI to CISPR 25 root-cause analysis

Projects

AI-Assisted EDA & CAD | *KiCad, FreeCAD, Claude Code* 2026 – Present

- Connecting KiCad and FreeCAD to an AI agent to explore AI-assisted schematic, PCB, and mechanical design

Education

University of Minnesota, Twin Cities Minneapolis, MN

Bachelor of Electrical Engineering (B.E.E.); GPA: 3.66/4.00 *Sep. 2018 – Dec. 2021*

- **Relevant Coursework:** Embedded System Design, Microcontrollers, Analog Electronics, Digital Design & Programmable Logic, Computer Architecture, Digital Signal Processing, Communication Systems, Transmission Lines & Fields, Senior Design Project
- **Honors:** Dean's List (Fall 2018, Spring 2019)

Minneapolis Community and Technical College Minneapolis, MN

Coursework in Mathematics; transferred to University of Minnesota *Jan. 2017 – May 2018*

Kyung Hee University Yongin, South Korea

Coursework in Electronics and Radio Engineering *Mar. 2013 – Dec. 2013*

Military Service

Republic of Korea Marine Corps Jun. 2014 – Mar. 2016

Sergeant, Communications; Honorable Discharge *South Korea*

Skills

Hardware Design: Schematic & PCB Design, Power Design for Compact Devices, SoC/MCU Bring-up, Wi-Fi/BLE Module Integration, Thermal Design
EMC & Reliability: CISPR 25 (RE/CE), ESD, JASO Environmental Testing, KC Certification, DFM
Tools & AI: OrCAD Capture, PADS Layout, KiCad, FreeCAD, Claude Code, Codex
Academic: Verilog, C (university coursework)
Languages: Korean (Native), English (U.S. Bachelor's Degree)